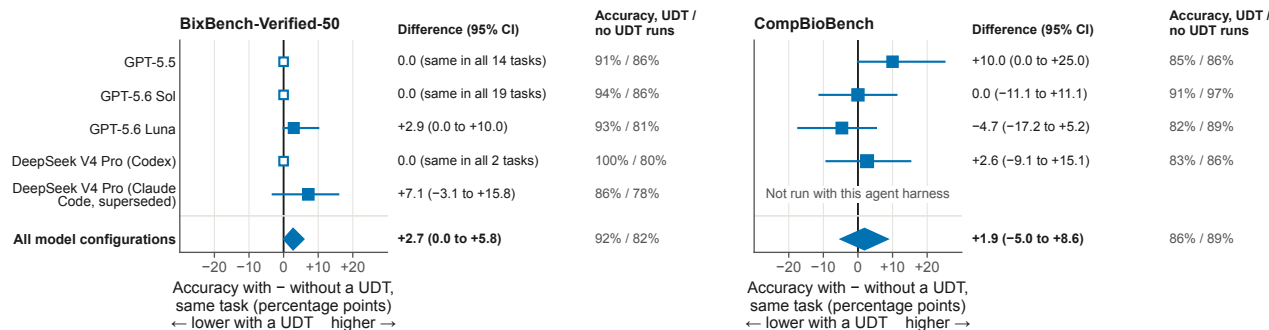
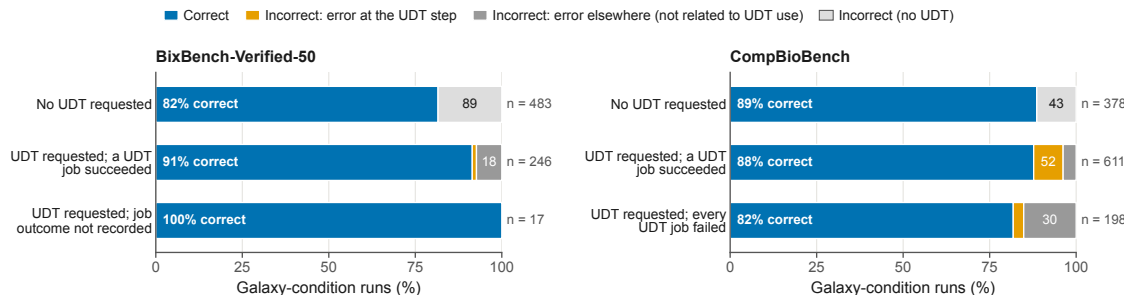


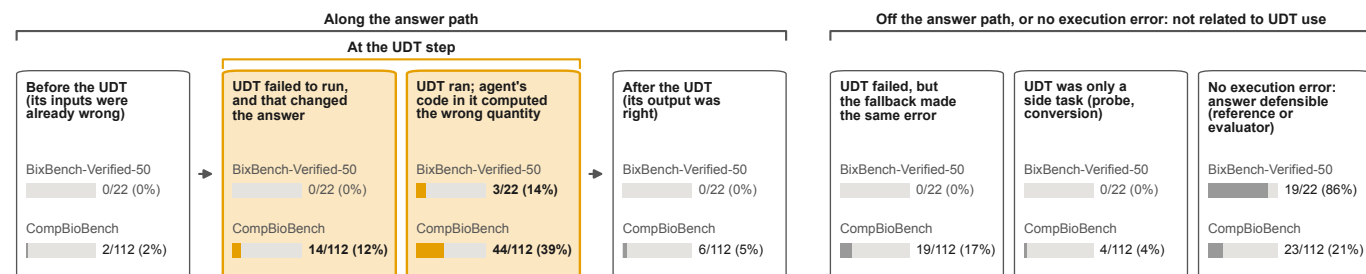
a Requesting a UDT was not associated with lower accuracy once the task is held fixed: pooled difference +2.7 points (0.0 to +5.8; BixBench-Verified-50) and +1.9 points (-5.0 to +8.6; CompBioBench)



b Runs whose UDT jobs succeeded were at least as accurate as runs without a UDT (91% versus 82% for BixBench-Verified-50; 88% versus 89% for CompBioBench); when every UDT job failed, most runs still reached a correct answer by another route (82% of 240 CompBioBench runs)



c In scored-incorrect UDT runs the error lay at the UDT step in 14% (BixBench-Verified-50) and 52% (CompBioBench); most of those were the agent's analysis written into the UDT, and the UDT mechanism itself failed in 14 of 134



Galaxy condition only; UDTs (user-defined tools: agent-written code that Galaxy runs as a job) were not offered for IWC tasks (0 of 120 runs). Runs without an execution trace are excluded (12 CompBioBench). a, Mantel-Haenszel risk difference within task (each model configuration; square size grows with the number of informative tasks) or within task × model configuration (diamond); 95% percentile cluster bootstrap (source capsules for BixBench-Verified-50, tasks for CompBioBench; 20,000 resamples). b, UDT job outcome from the archived analysis history (rows with fewer than 5 runs are in Source Data only: BixBench-Verified-50, every UDT job failed, 4 runs; CompBioBench, outcome not recorded, 1 run); error location from c. c, Trace-level audit of all 134 scored-incorrect UDT runs against a fixed codebook (on_demand/udt_audit/; layout after the stage-ordered failure taxonomy of Cemri et al., 2025).